

Teacher Academic MIS

Deep-Dive Code Audit & Production-Readiness Report

Stack: React 18 + Vite 5 + TypeScript (strict) + Tailwind v3 + Supabase (Postgres/Auth/RLS) + Cloudflare Pages
Scope: 209 source files • 57 SQL migrations • 2 edge functions
Date: 2026-08-17 | Branch: feature/superhuman-quiz-design

VERDICT: NOT PRODUCTION-READY YET

Core quiz, auth, routing aur database security (RLS + edge functions) **solid** hain. Lekin 5 blockers hain: AttendanceView pe **@ts-nocheck** (poora type-safety off), koi **ErrorBoundary** nahi (crash = white screen), **3 routing tests fail**, RosterView + TeacherManagementView mein **fake/simulated buttons**, aur **146 hardcoded colors** (dark-mode inconsistency). Inhe fix kiye bina deploy risky hai.

CRITICAL 3	HIGH 5	MEDIUM 8	LOW 5
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1. Audit Coverage

Ye report un files ke deep reading pe based hai jo directly padhi gayi. Kuch modules 'sampled' hain (pattern-grep se signal, spot reads) — clearly marked.

Module / Area	Depth
Quiz (attempt / access / creation-data / parsers)	Full read
Auth + Routing (App.tsx, AuthContext, SignInRoute)	Full read
Database RLS (0002, 0014) + is_teacher/owner isolation	Full read
Cloudflare edge functions (generate-quiz, admin-create-teacher)	Full read
Shared UI kit (foundation/forms/feedback/OtpInput/ThemeToggle)	Full read
Data-layer error handling (support.ts)	Full read
TeacherManagementView, RosterView, AttendanceView (header)	Full read
SelectedSectionContext, theme, tailwind config, index.css	Full read
Marks/Analytics/Leaderboard/Heatmap/Dashboard	Sampled (grep + spot)
Material/Assignment/Syllabus/Onboarding internals	Sampled (grep + spot)
Remaining 50 migrations (0001,0003-0056)	Sampled (index + key ones)

Note: Pehla plan tha 8 parallel audit-agents chalane ka; wo sab token-quota exhaust hone se fail ho gaye, isliye audit manually kiya gaya. 'Sampled' areas ki confirmed findings neeche hain, par unmein aur bhi ho sakte hain — full pass baaki hai.

2. Critical Findings

ID	File	Issue & Impact	Fix
C1	views/AttendanceView.tsx:1	File pe // @ts-nocheck hai — poori attendance view mein TypeScript kuch check nahi karta. Null access / wrong prop / undefined method silently ship. Attendance core feature hai; crash = classes mark nahi hongii. (fix.cjs ne inject kiya.)	@ts-nocheck hatao; real type errors fix karo (~32 hardcoded colors bhi saath).
C2	presentation/App.tsx	Poore app mein koi React ErrorBoundary nahi. Kisi bhi component ka render crash = pura app white screen, koi recovery nahi.	Root pe <ErrorBoundary> with fallback UI wrap karo.
C3	App.routing.test.tsx	3 routing tests fail. Test window.location.replace('/') expect karta hai par onSignedIn jaan-boojh ke navigate nahi karta (redirect SignInRoute re-render se hota hai). Test vs code mismatch — sign-in redirect regression ab catch nahi hoga.	Test ko React-Router Navigate ke hisab se update karo (pathname assert).

3. High Findings

ID	File	Issue & Impact	Fix
H1	views/QuizAttemptView.tsx	Double-submit race: handleSubmit + handleAutoSubmit dono submitAttempt call karte hain. Timer 0 pe user 'Submit' dabaye to dono fire — koi submittedRef guard nahi.	Ek submittedRef/isSubmitting guard dono paths pe.
H2	views/RosterView.tsx	Fake/simulated buttons: dummyAction() — 'Add Student', 'Assign Section', 'Delete Students', 'Email Student', 'Suspend Account', 'View Full Record' sab sirf toast dikhate hain, kuch save nahi hota. Teacher ko real lagega.	Real backend se wire karo, ya 'Coming soon / Demo' clearly label karo.
H3	views/TeacherManagementView.tsx	Same as H2 — dummyAction(), 'Simulated import/assignment', hardcoded fake profile data (Total Classes, subjects). Admin ko misleading.	Wire ya clearly disable/label karo.
H4	data/access/support.ts + domain/*	Error-handling do philosophy: data-access layer THROWS DataAccessError; domain (fileStorage, inputGuard) returns Result<T,E>; RPC parsers tagged-union {status,reason}. Ek component ko try/catch AND .ok dono handle karne padte hain.	Ek convention decide karo (recommend: data layer bhi Result<T>).
H5	data/access/support.ts:46	unwrap() success pe null deta hai (0 rows), error pe throw. Caller har successful single-read pe null-check bhoole to 'not found' crash ban jata hai.	unwrapSingle() add karo jo null pe throw kare.

4. Medium Findings

ID	File	Issue & Impact	Fix
M1	views/*.tsx (11 files)	146 hardcoded colors (hex / emerald- / indigo- / slate-) design tokens ki jagah. AttendanceView 32, StudentQuizAccessView 72, DashboardView 11, MarksCalculatorView 7. Dark-mode toota / theming inconsistent.	status-*/accent/surface/text tokens use karo.

ID	File	Issue & Impact	Fix
M2	data/supabase/client.ts:57-63	Env missing hone pe placeholder (localhost:54321, placeholder-anon-key) pe fallback + sirf console.warn. Prod mein galat deploy chup-chaap boot hoga, har backend call fail.	PROD mein descriptive error throw karo, placeholder nahi.
M3	QuizCreationView.tsx:101, fileStorage.ts:125, localDemoMode.ts:171	Share/file token = Date.now()+Math.random().toString(36) — low-entropy, guessable. Quiz link enumeration risk (enrollment/OTP gate se mitigated).	crypto.randomUUID() / crypto.getRandomValues use karo.
M4	views/OtpInput.tsx	Purani purple palette hardcoded (#15157d/#818cf8/#0f172a). TeacherSignIn + StudentQuiz dono mein use, design system se mismatch, dark-mode inconsistent.	Design tokens (accent/surface/border) pe migrate karo.
M5	views/QuizAttemptView.tsx (EOF)	Inline cx() helper banaya jabki ui/utils.ts mein cx() already export hai. Duplication.	Shared cx import karo.
M6	views/QuizAttemptView.tsx	Timer pe animate-pulse; prefers-reduced-motion guard nahi (Tailwind default pulse index.css ki reduced-motion list mein nahi). Motion-sensitivity WCAG issue.	motion-reduce:animate-no ne add karo.
M7	context/SelectedSectionContext.tsx:114	loadSections() catch khaali — network fail pe sections silently empty, user ko koi error nahi, bas blank dropdowns.	Error state surface karo + retry.
M8	data/migrations/	Migration numbering collision: do 0018 (dedupe_syllabus_subjects + syllabus_master) aur do 0021 (quiz_share_token + sem5_electives). Apply order ambiguous.	Renumber duplicates to unique sequence.

5. Low Findings

ID	File	Issue & Impact	Fix
L1	views/QuizAttemptView.tsx	Question palette chips h-6 w-6 (24px) — touch target chhota (min 44dp recommended).	Chip size badhao ya hit-area padding do.
L2	auth/AuthContext.tsx	get_my_role() har auth-state-change pe call, koi caching nahi — multi-tab pe redundant RPC.	Role result short-cache karo.
L3	repo root	Untracked fix.cjs + update-ui.cjs — throwaway codemods; fix.cjs hi @ts-nocheck inject karta hai (C1 ki jad).	Delete karo ya scripts/ mein move + gitignore.
L4	views/AttendanceView.tsx:1589	confirmButtonRef 'as any' cast (@ts-nocheck ke upar) — ref typing bypass.	Proper ref type do.
L5	views/AttendanceView.tsx	Hardcoded emerald-500 focus/bg colors design tokens ki jagah (C1/M1 ka hissa).	accent/status-green tokens.

6. Security Assessment — Strong

Security posture is the project's strongest area. Jo cheezein sahi hain:

Area	Finding
RLS coverage	0002 har table pe RLS enable + default-DENY; teacher & student-own-row policies (OR-combined).
Per-teacher isolation	0014 owner_id = auth.uid() based RLS operational tables pe; ek teacher doosre ka data nahi dekh sakta.
Answer-key safety	questions.correct_index teacher-only RLS; grading server-side SECURITY DEFINER RPC se; client ko keys nahi milte.
Edge fn: generate-quiz	GEMINI_API_KEY server-only; caller ka JWT-scoped client + is_teacher() RPC verify hota hai privileged call se pehle.
Edge fn: admin-create-teacher	SERVICE_ROLE_KEY server-only; is_admin() RPC verify ke BAAD hi service-role client banta hai; client claim trust nahi.
Teacher/student separation	0027 DB trigger student ko teacher-escalate hone se rokta hai (client bypass-proof).
Parameterized access	Sara data access Supabase query-builder / .rpc() se; koi ad-hoc string SQL nahi — SQL-injection surface minimal.

Minor security gaps: generate-quiz CORS 'access-control-allow-origin: *' (auth-gated, low risk); OTP endpoints pe sirf 60s client cooldown (server rate-limit Supabase pe rely); localStorage mein email plaintext (rememberMe).

7. Known Functional Bug (from PROJECT_HANDOFF)

Quiz wrong-section: request_quiz_access (migration 0040) student upsert mein section_id set nahi karta. Quiz-link se self-register hone wale students ka section_id = NULL reh jata hai, jisse har section-restricted quiz unhe 'wrong section' deny karta hai. Abhi manual SQL se fix hota hai; proper migration pending.

8. Production-Readiness Checklist

Item	Status
Database RLS + authorization	READY
Edge function security (secrets, auth-gating)	READY
Core quiz flow (access -> attempt -> submit -> review)	READY
Domain logic unit tests (312 pass)	READY
TypeScript type-safety across all files	BLOCKED (C1: @ts-nocheck)
Crash resilience (ErrorBoundary)	BLOCKED (C2: none)
Routing regression tests green	BLOCKED (C3: 3 fail)
No fake/placeholder UI shown as real	BLOCKED (H2/H3)
Consistent dark-mode / design tokens	GAP (M1: 146 colors)
Error monitoring (Sentry/logging)	MISSING
E2E tests (sign-in, quiz)	MISSING
Security headers (CSP/X-Frame)	MISSING
CI/CD pipeline	MISSING

9. Recommended Action Plan

Aaj / deploy se pehle (blockers):

- C1 — AttendanceView se @ts-nocheck hatao, type errors fix karo.
- C2 — Root ErrorBoundary add karo.
- C3 — 3 routing tests fix karo (Navigate-based assert).
- H1 — Quiz double-submit guard.
- H2/H3 — RosterView + TeacherManagementView ke fake buttons wire ya clearly label karo.

Is hafte (high):

- H4/H5 — Unified Result<T> error model; unwrapSingle().
- M2 — Prod env-missing pe fail-fast.
- M3 — crypto-based token generation.
- M8 — Duplicate migration numbers renumber.
- Sampled modules (Marks/Analytics/Roster/Material/Onboarding) ka full pass.

Agla sprint (hardening):

- M1/M4 — 146 hardcoded colors -> tokens; OtpInput palette.
- Error monitoring (Sentry) + structured logging.
- E2E tests (Playwright): sign-in, quiz attempt+submit.
- Security headers (CSP) + CI/CD pipeline.
- Quiz wrong-section proper migration fix (section_id on upsert).

Report generated from direct source reading. 'Sampled' modules ki findings confirmed hain par exhaustive nahi — un areas ka full deep pass abhi baaki hai.